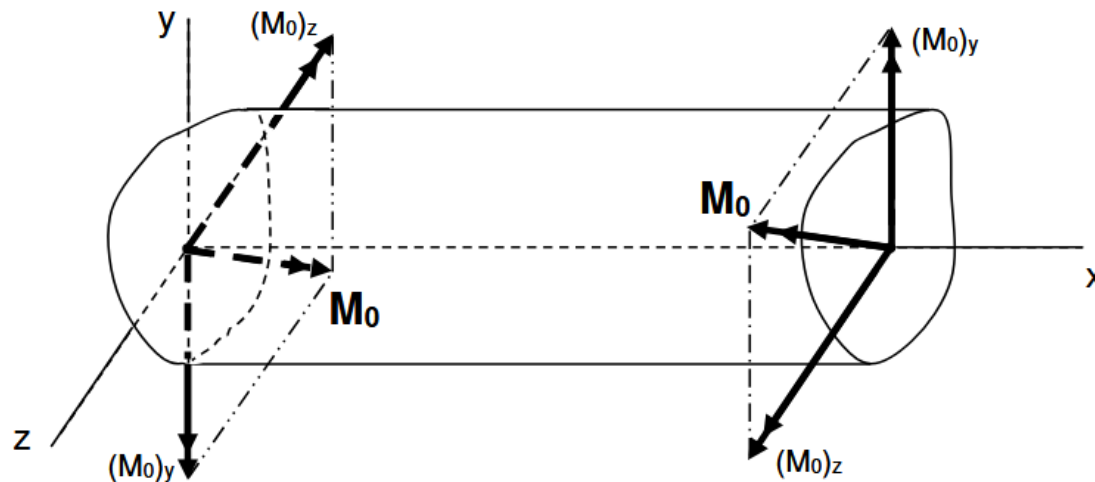


4.1: General Flexure Formula

- Recall the equation $\sigma_x = \frac{My}{I}$ for bending-induced normal stress
- We can write this equation more specifically as $\sigma_x = -\frac{M_z y}{I_z}$:
 - M_z is the applied moment about the z axis
 - y is the distance of the differential area element above or below the neutral axis (which is in the z direction)
 - I_z is the area moment of inertia of the cross-section about the z axis (effectively a measure of the beam's resistance to rotation about the z axis)
- However, this is a special case of a more general equation; this simplified version applies under the following assumptions:
 - The applied moment (due to either transverse load or concentrated couple) is strictly about the z axis
 - Cross-section symmetric in y or z (i.e. the cross-section is such that either the y or z axis is a line of symmetry)
- Here we will derive the more general equation ----->

4.1: General Flexure Formula

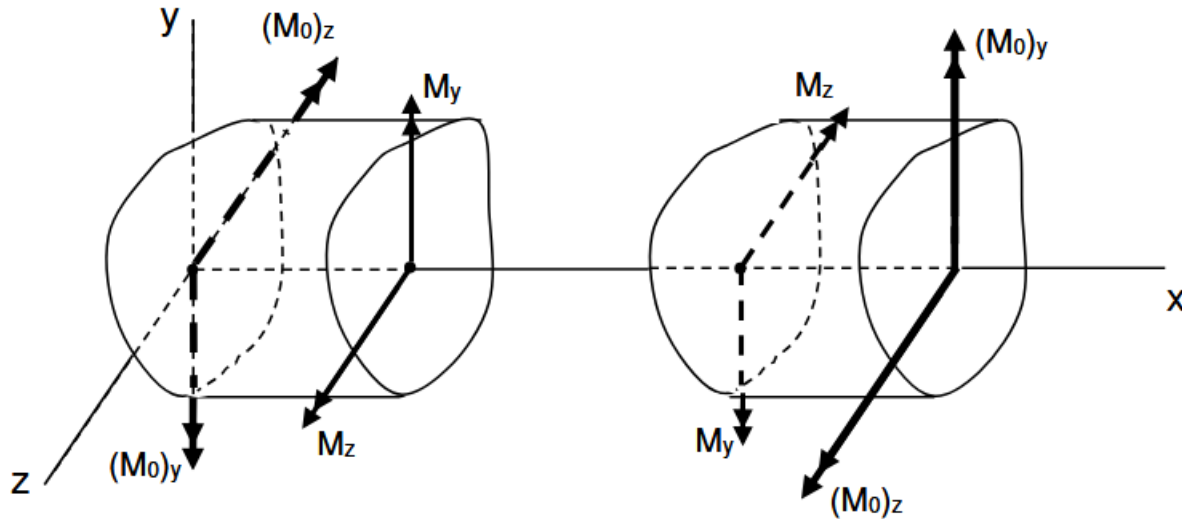
- Consider the beam of arbitrary-shaped cross-section below, with the x axis along the length of the beam as usual
- The beam is experiencing pure bending (i.e. no axial force and no torsion)



- Here we see two equal and opposite concentrated couples $\bar{\mathbf{M}}_0$
 - One can be thought of as being applied to the beam, and the other as a reaction due to a support at the end
 - $\bar{\mathbf{M}}_0$ is not constrained to lie along the z axis, but rather can lie anywhere in the y-z plane (components M_{0y} and M_{0z} are shown)

4.1: General Flexure Formula

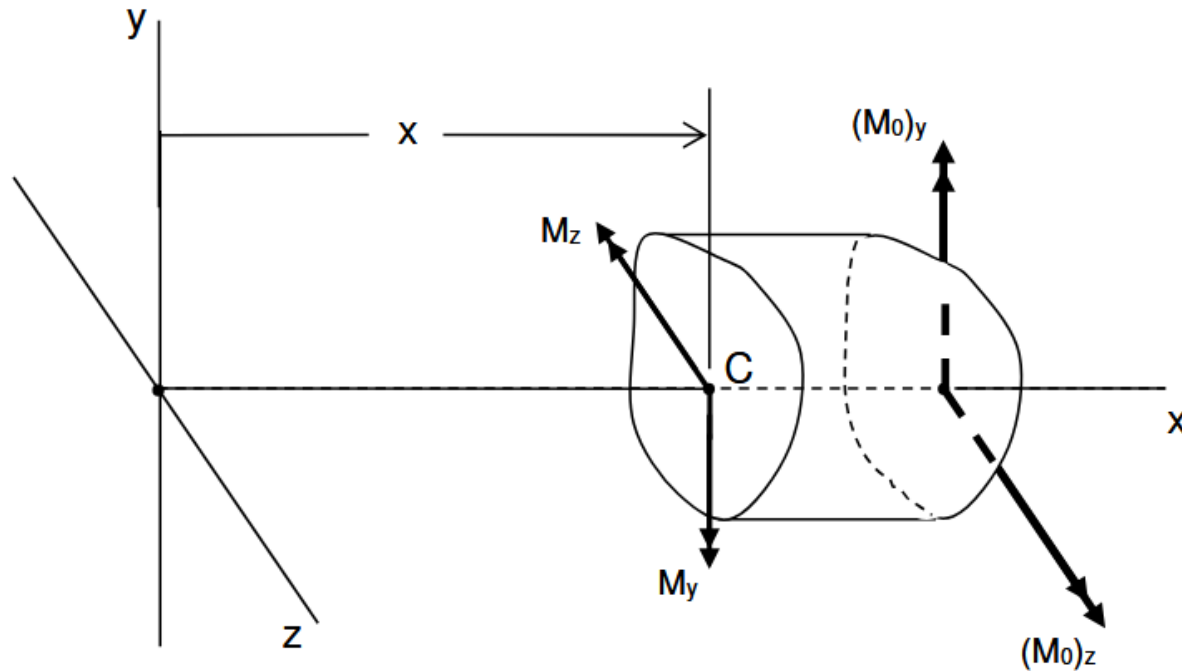
- If we cut the beam somewhere along its length, this exposes the internal moment on either side of the cut:



- On both sides of the cut, note that the components M_y and M_z of the internal moment are opposite (i.e. negative) to that of the components at the end of the beam
- That is, $M_y = -M_{0y}$ and $M_z = -M_{0z}$

4.1: General Flexure Formula

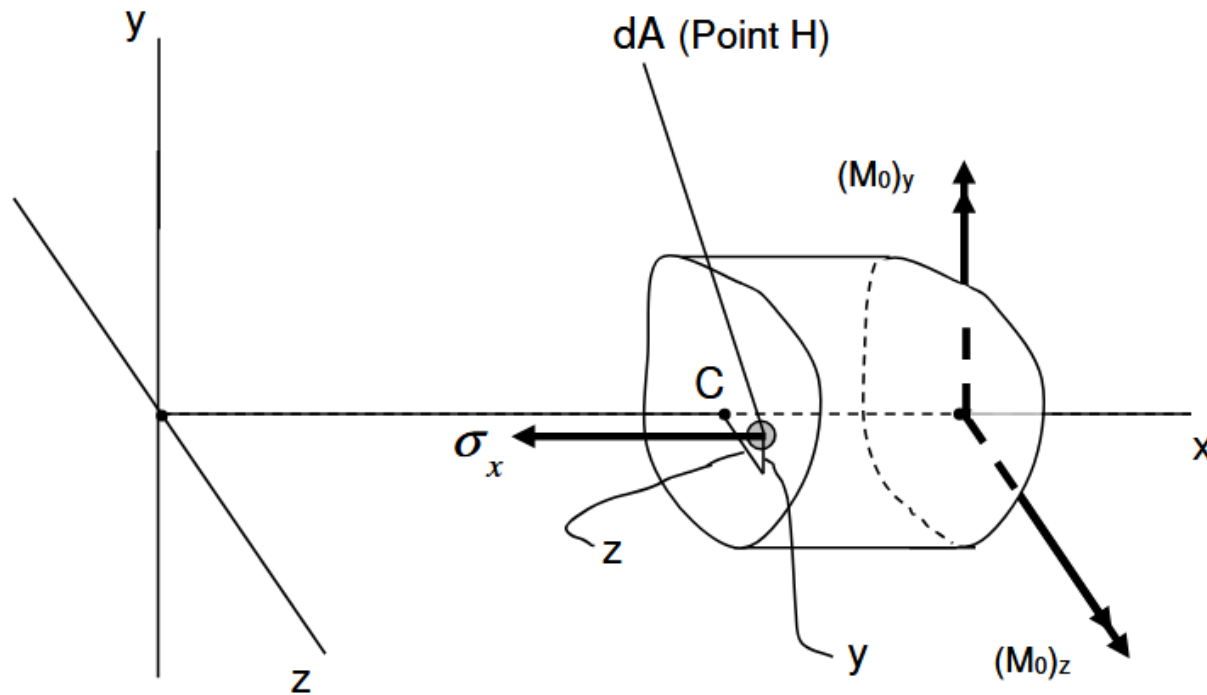
- Let's focus on the right portion of the beam, with the point C shown along the x axis (i.e. where the x axis pierces the interior cross-section)



- We know that M_y and M_z result in an infinite number of normal stresses distributed in some fashion over the interior cross-section, such that $M_y = \int_A z \sigma_x dA$ and $M_z = - \int_A y \sigma_x dA$

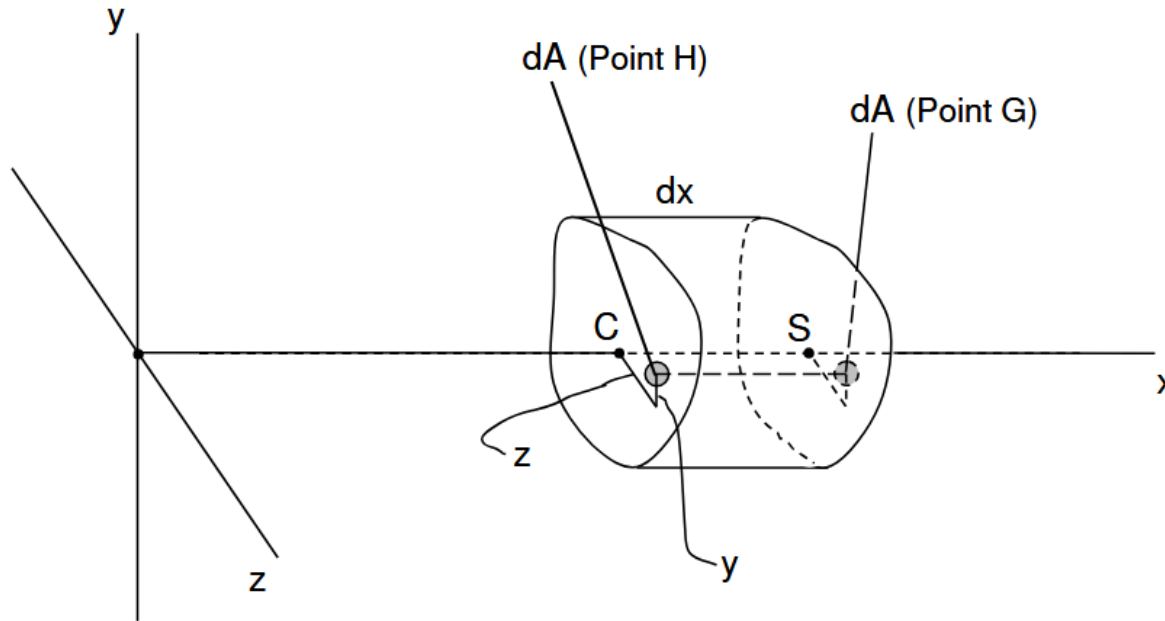
4.1: General Flexure Formula

- One of these normal stresses is shown below, acting at point H located at a y-distance and a z-distance from point C:



4.1: General Flexure Formula

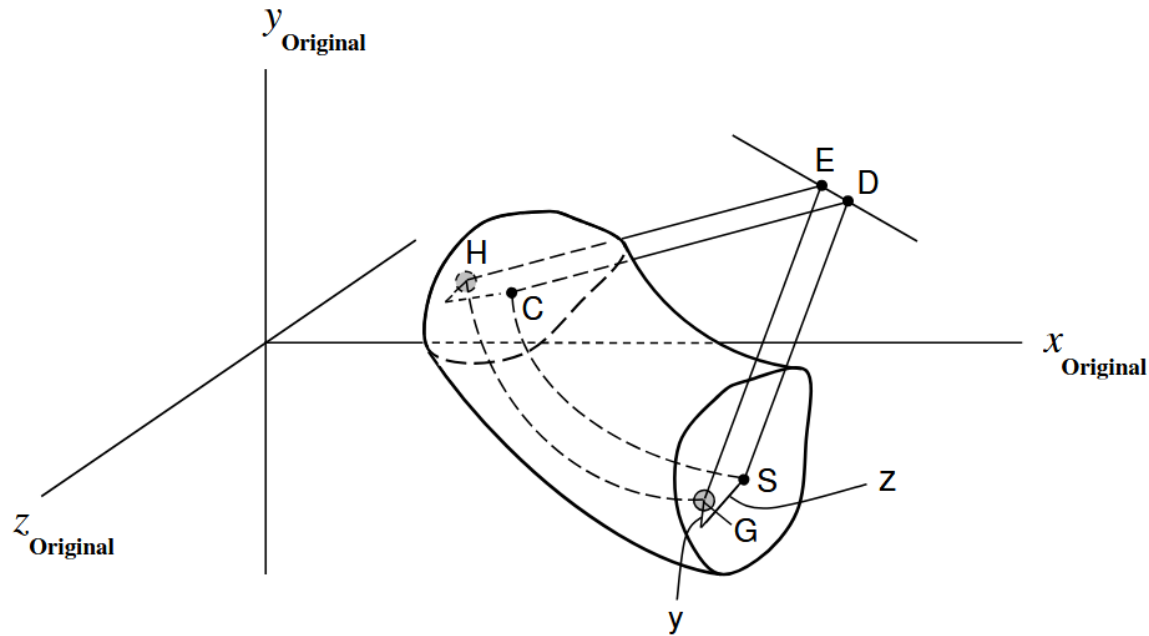
- Now consider a section of the beam of infinitesimally small thickness dx :



- Points S and G on the right side of the section are analogous to points C and H on the left side of the section; thus line $\overline{CS}$ is along the x axis and line $\overline{HG}$ is parallel to the x axis
- So line $\overline{CS}$ is above (positive y) and to the side (positive z) of line $\overline{HG}$

4.1: General Flexure Formula

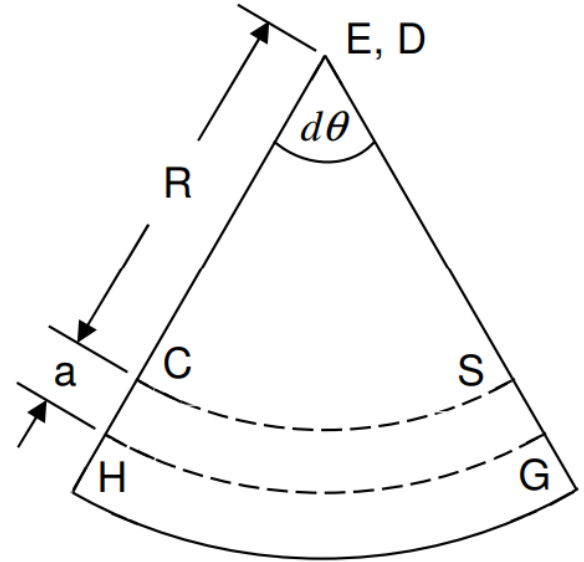
- Below is the same section after deformation resulting from the bending; the lines $\overline{CS}$ and $\overline{HG}$ will each bend into an arc



- Points D and E represent the **center of curvature** of arcs $\overline{CS}$ and $\overline{HG}$, respectively; note that if we connect each end of an arc with its center of curvature, it resembles a “pie slice”
- An artifact of pure bending is that the centers of curvature of all fibers in the section lie along the same line

4.1: General Flexure Formula

- If we view the beam along this line of common centers and focus on pie slice DCS and pie slice EHG, it looks like this:



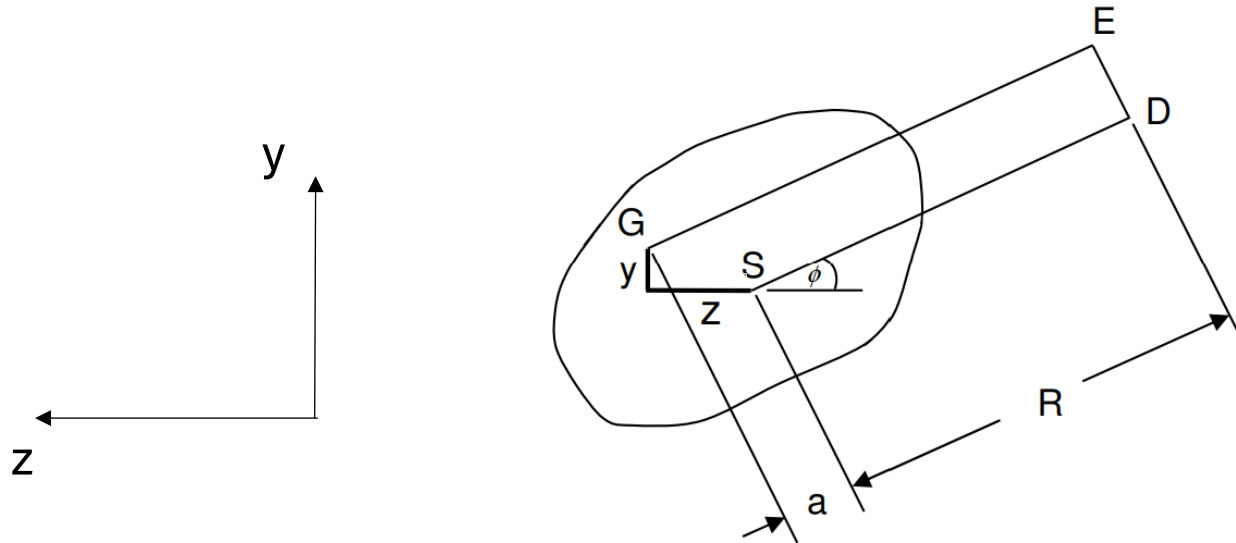
- Here we label the **radius of curvature** of arc $\overline{CS}$ as “ R ” and the ROC of arc $\overline{HG}$ as “ $R + a$ ”
- (NOTE: the 3rd arc shown below arc HG really doesn’t add anything to the figure, **ignore it**)

4.1: General Flexure Formula

- Recall that Hooke's Law yields $\sigma_x = E\varepsilon_x = E\frac{\delta}{L}$
- We see from the figures above that the length of line $\overline{HG}$ before deformation is dx and the length after deformation is $(R + a)d\theta$; similarly, the length of line $\overline{CS}$ before deformation is dx and the length after deformation is $Rd\theta$
- Applying Hooke's Law to line $\overline{HG}$ yields $\sigma_x = E\frac{(R + a)d\theta - dx}{dx}$
- Let us assume that line $\overline{CS}$ undergoes no deformation (we will explore this later); therefore $dx = Rd\theta$
- So the equation above becomes $\sigma_x = E\frac{(R + a)d\theta - Rd\theta}{Rd\theta} \rightarrow \sigma_x = E\frac{a}{R}$

4.1: General Flexure Formula

- **Let's look at the right face of this section from the "y-z plane" perspective (i.e. along the x axis):**



- Here ϕ is the angle each pie slice makes with the y-z plane
- If we stare at this figure long enough, we see that $a = z \cos \phi - y \sin \phi$
- Substituting into the previous equation yields $\sigma_x = E \left[-\frac{\sin \phi}{R} y + \frac{\cos \phi}{R} z \right]$

4.1: General Flexure Formula

- If we define constants $B_1 = \frac{\sin\phi}{R}$ and $B_2 = \frac{\cos\phi}{R}$, then the previous equation yields $\sigma_x = -EB_1y + EB_2z$
- For a given bending scenario (i.e. a given M_y and M_z applied to a beam with a given cross-section), can we solve B_1 and B_2 ?
- Recall that $M_y = \int_A z\sigma_x dA$; substituting the above equation for σ_x yields $M_y = \int_A z(-EB_1y + EB_2z) dA = EB_2 \int_A z^2 dA - EB_1 \int_A yz dA$
- Recognize that:
 - $\int_A z^2 dA$ is the **area moment of inertia** of the cross-section about the y axis, which we label I_y
 - $\int_A yz dA$ is the **area product of inertia** of the cross-section about the y and z axes, which we label I_{yz}

4.1: General Flexure Formula

- So we can rewrite the previous equation as $M_y = EB_2I_y - EB_1I_{yz}$
- Similarly, $M_z = -\int_A y\sigma_x dA = -\int_A y(-EB_1y + EB_2z)dA = EB_1\int_A y^2 dA - EB_2\int_A yz dA = EB_1I_z - EB_2I_{yz}$
- So for a given scenario, M_y , M_z , I_y , I_z , and I_{yz} will be known and the above 2 equations can be solved for B_1 and B_2 :

$$B_1 = \frac{I_y M_z + I_{yz} M_y}{E(I_y I_z - I_{yz}^2)}, \quad B_2 = \frac{I_{yz} M_z + I_z M_y}{E(I_y I_z - I_{yz}^2)}$$

- Inserting these expressions into $\sigma_x = -EB_1y + EB_2z$ yields

$$\sigma_x = \frac{-y(I_y M_z + I_{yz} M_y) + z(I_{yz} M_z + I_z M_y)}{I_y I_z - I_{yz}^2}$$

This is known as the “General Flexure Formula”